

B.E.S DEGREE COLLEGE

DEPARTMENT OF COMPUTER APPLICATION



Affiliated to Dr. Manmohan Singh Bengaluru City University

Accredited by NAAC with B++ Grade

Synopsis

Chronos - Bengaluru Urban Delivery Analytics Dashboard

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Project report submitted in partial fulfillment of the requirements of sixth semester , BES degree College .

Project Title: Chronos - Bengaluru Urban Delivery Analytics Dashboard

Main Objective: To build a data analytics web dashboard that visualizes and analyzes the statistical impact of local traffic density, weather conditions, and time of day on urban food delivery times in Bengaluru.

Synopsis:

- Chronos is a data visualization Web App built using the MERN stack (MongoDB, Express, React, Node.js) and Python.
- The project utilizes urban food delivery datasets (like Swiggy/Zomato Kaggle data) to track the friction of city logistics.
- Python (Pandas) is used to clean and process the raw CSV data to find correlations between weather, time, and delivery delays.
- The React frontend utilizes charting libraries (like Chart.js or Recharts) to display interactive graphs, such as comparing average delivery delays in Bommanahalli at 6 PM versus 9 PM.
- Processed analytical data is stored in MongoDB for fast retrieval and rendering on the user dashboard.

Requirements:

Hardware: Standard PC/laptop with minimum 4GB RAM, 1.5 GHz processor, 10GB free space.

Software: Windows 10/11 or Linux, VS Code, Google Chrome, Node.js, Python 3.x, MongoDB Atlas Account.

Language:

- Python (Data Cleaning & Processing)
- JavaScript / TypeScript (Logic and API integration)
- HTML5 & CSS3 (Structure and Design)

Backend:

- Node.js & Express.js (Server & API Routing)
- MongoDB Atlas (Database)

Frontend:

- React.js (User Interface)
- Tailwind CSS (Responsive Design)
- Chart.js / Recharts (Data Visualization)

Connectivity:

- **RESTful APIs**
- **Browser-based visualization (no app install)**
- **Cloud Database Connection (MongoDB URI)**

Team Members: Lakshmi and Vinodhini R BCA 6th Semester